

Decision Transformer Warm-Start for Minimum-Time Vehicle Trajectory Optimization

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Final Project Presentation

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Problem

- Minimum-time racing line optimization is a constrained planning problem near the tire friction boundary.
- The planner must satisfy nonlinear vehicle dynamics, hard track limits, and static obstacle-clearance constraints.
- The project goal is to test whether an offline sequence model can provide a better warm start for this nonlinear program.
- The target benefit is lower solver iterations and lower wall-clock solve time without sacrificing feasibility or lap time quality.

Setup

- Stage 1: solve the minimum-time trajectory optimization problem offline to generate expert trajectories.
- Stage 2: train a Decision Transformer (DT) to predict warm-start trajectories $(X_{\text{init}}, U_{\text{init}})$ for the same downstream optimizer.



Background

- The planner is a minimum-time nonlinear program in Frenet coordinates solved by spatial direct collocation.
- In implementation, the nonlinear program is built in CasADi and solved with FATROP.
- Vehicle model: nonlinear single-track model configured to the 2018/2019 VW Golf GTI from prior Stanford autonomous-racing work [1].
- The optimizer minimizes lap time while enforcing nonlinear dynamics, track limits, forward progress, and obstacle clearance.
- For learning, Decision Transformer is used as an offline RL sequence model conditioned on return-to-go, state, and action history [2].
- The approach is motivated by prior work on

Optimization Problem

$$\min_{x_{0:N}, u_{0:N}} t_N + \lambda_u \sum_{k=0}^{N-1} \|u_{k+1} - u_k\|^2$$

$$x_{k+1} = x_k + \frac{\Delta s}{2} \left(f_s(x_k, u_k) + f_s(x_{k+1}, u_{k+1}) \right)$$

$$e_{\min}(s_k) \leq e_k \leq e_{\max}(s_k), \quad \dot{s}_k \geq \epsilon_s$$

$$1 - \kappa(s_k)e_k \geq \epsilon_\kappa, \quad F_{x,\min} \leq F_{x,k} \leq F_{x,\max}$$

$$\|p(s_k, e_k) - p_{\text{obs},i}\|^2 \geq R_{\text{safe},i}^2$$



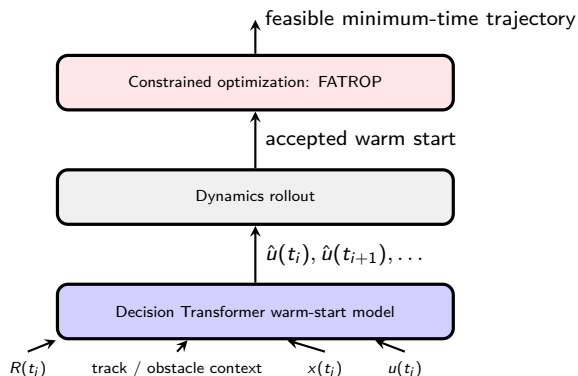


Figure 1: Offline warm-start pipeline.

$$\mathcal{L}_k = \|u_k - \hat{u}_k\|_2^2 + \lambda_x \|x_{k+1}^{\text{obs}} - \hat{x}_{k+1}^{\text{obs}}\|_2^2$$

- Input: return-to-go, state, track, and obstacle context.
- Output: warm-start controls and rolled-out states.
- DT: context length 50, 6 layers, 4 heads, 192-dim embeddings.
- Loss: action imitation plus next-state consistency.
- Inference: autoregressive prediction with dynamics rollout.



Experiments

- Benchmark: Oval-first evaluation with two fixed evaluation sets. The Oval track is used because it matches the setting used in the earlier Stanford GTI trajectory-optimization work:
 - no-obstacle evaluation set,
 - obstacle evaluation set with 1–4 circular obstacles.
- Downstream solver: FATROP for expert generation, repair segments, and warm-started solves.
- Dataset mix:
 - 31,710 shifted accepted full laps,
 - 400 hard repairs,
 - 1,000 post-projection repairs,
 - 300 failure-conditioned repairs.

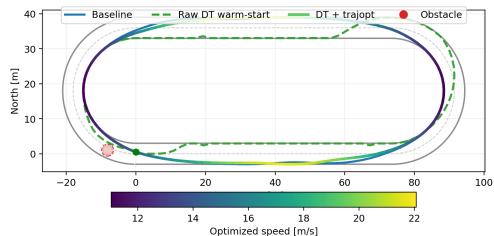


Figure 2: Representative obstacle case from the corrected evaluation pipeline.



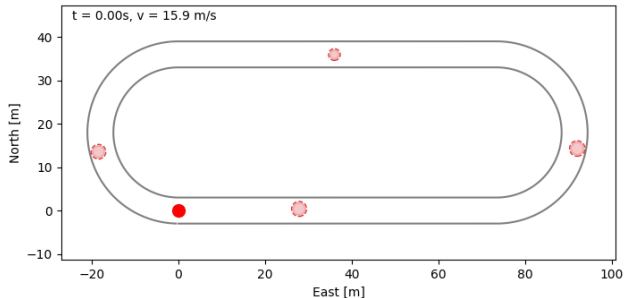
Trajectory Animation

This slide links to the full trajectory GIF used during optimization debugging.

Asset

- FATROP obstacle-track animation
- Oval track with obstacles
- horizon $N = 120$

[Open trajectory GIF](#)



Results and Interpretation

- The project started from a parent DT trained on nominal Oval trajectories and then improved the warm-start pipeline through targeted training and debugging.
- In later runs, raw DT trajectories became more structured, handled obstacles more plausibly, and in some cases slightly reduced solver-only time.
- This showed that the DT was learning useful trajectory structure, even though the full warm-start pipeline was not yet consistently faster end to end.

Main improvements during training and debugging

- larger-context DT training improved nominal trajectory quality,
- repair datasets exposed the model to harder rollout states,
- rollout and evaluation fixes made downstream behavior much easier to interpret.

Setting	WS accept	Baseline solve	DT total
No obstacles	20%	49.06s	51.98s
1–4 obstacles	20%	56.36s	60.79s



Results: Training and Qualitative Behavior

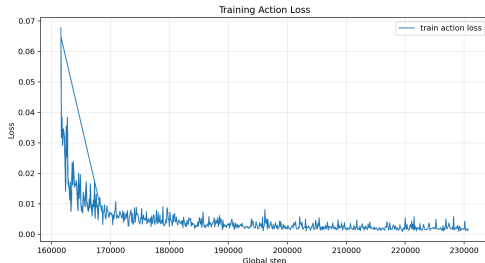


Figure 4: Later fine-tuning runs trained stably under the larger-context DT setup.

- The strongest result was not a full end-to-end speedup, but a more interpretable and more structured warm-start behavior.
- Across stricter evaluation settings, the same pattern remained:
 - plausible raw DT trajectories,
 - occasional solver-only benefit,
 - but not yet reliable end-to-end gains.
- A major project outcome was identifying the remaining bottleneck clearly enough to target the next iteration.



Conclusions and Next Steps

- The main contribution is a working framework for learned warm-starts in constrained racing trajectory optimization.
- The full pipeline now covers:
 - offline expert generation,
 - DT training and fine-tuning,
 - repair-data generation,
 - rollout diagnostics,
 - downstream benchmarking.
- The DT learned meaningful trajectory structure and produced real warm-start signal.
- The harder lesson was that deployment metrics matter more than offline loss alone.

Next step: continue training on the difficult rollout states revealed by diagnostics so that improved raw DT trajectory quality turns into more consistent optimizer gains.



- ① J. K. Subosits and J. C. Gerdes, "Impacts of Model Fidelity on Trajectory Optimization for Autonomous Vehicles in Extreme Maneuvers," *IEEE Transactions on Intelligent Vehicles*, 2021.
- ② L. Chen et al., "Decision Transformer: Reinforcement Learning via Sequence Modeling," *Advances in Neural Information Processing Systems*, 2021.
- ③ T. Guffanti and S. D'Amico, "Transformers for Trajectory Optimization with Application to Spacecraft Rendezvous," *IEEE Aerospace Conference*, 2024.

